

Secondary Examination, 2018

Mathematics

Time- $3\frac{1}{4}$ Hours

M.M- 80

GENERAL INSTRUCTIONS TO THE EXAMINEES:

- 1) Candidate must write first his/ her Roll No. on the question paper compulsorily.
- 2) All the questions are compulsory.
- 3) Write the answer to each question in the given answer- book only.
- 4) For questions having more than one part the answers to those parts are to be written together in continuity.
- 5) If there is any error/difference/contradiction in Hindi and English versions of the question paper of Hindi version be treated valid.
- 6)

Section	Q.Nos.	Marks per question
A	1-10	1
B	11-15	2
C	16-25	3
D	26-30	6
- 7) There are internal choices in Q. Nos. 28 and 30.
- 8) Write on both sides of the pages of your answer-book. If any rough work is to be done, do it on last pages of the answer-book and cross with slant lines and write 'Rough Work' on them.
- 9) Draw the graph of Q. No. 26 on the graph paper.

PART-A

Question.1 Find the value of $31\frac{1}{6} \times 31\frac{5}{6}$ by using Ekaadhiken Purven Sutra.

Solution.

Simplify the expression,

$$\begin{aligned} & 31\frac{1}{6} \times 31\frac{5}{6} \\ & 31 \times (31+1) \times \frac{1}{6} \times \frac{5}{6} \\ & 31 \times 32 \times \frac{5}{6} \\ & = 992\frac{5}{36} \end{aligned}$$

Hence, the value is $992\frac{5}{36}$

Question.2 Solve $\frac{1}{x-3} + \frac{1}{x-7} = \frac{1}{x-1} + \frac{1}{x-9}$

Solution.

Simplify the expression,

$$= \frac{1}{x-3} + \frac{1}{x-7} = \frac{1}{x-1} + \frac{1}{x-9}$$

add in denominator

$$x-3+x-7 = x-1+x-9$$

$$2x-10 = 2x-10$$

$$2x-10=0$$

$$2x=10$$

$$x=5$$

Hence, the value is 5.

Question.3 Write the sum of powers of prime factors of 196.

Solution.

Simplify the expression,

$$\begin{aligned} 196 &= 7 \times 2 \times 2 \times 7 \\ &= 2^2 \times 7^2 \\ \Rightarrow s(2+2) &= 4 \end{aligned}$$

Hence, the value of 4.

Question.4 Write the value of $\cos 50^\circ \cdot \operatorname{cosec} 40^\circ$

Solution.

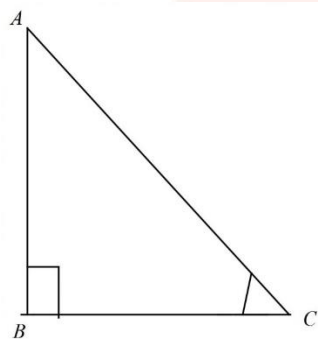
Simplify the expression,

$$\begin{aligned} &\cos 50^\circ \cdot \operatorname{cosec} 40^\circ \\ &= \cos 50^\circ \cdot \operatorname{cosec} (90^\circ - 50^\circ) \quad [\operatorname{cosec} (90^\circ - \theta) = \sec \theta] \\ &= \cos 50^\circ \cdot \sec 50^\circ \\ &= \cos 50^\circ \times \frac{1}{\cos 50^\circ} = 1 \\ &= 1 \end{aligned}$$

Hence, the value is 1.

Question.5 If the ratio of the length of a vertical bar to its shadow is $1:\sqrt{3}$, then find elevation angle of the sun.

Solution.



Simplify the expression,

Let AB be the length –post and BC be its shadow.

$\angle ACB = \theta = \text{angle of elevation of sun.}$

$$\text{now, } \tan \theta = \frac{AB}{BC}$$

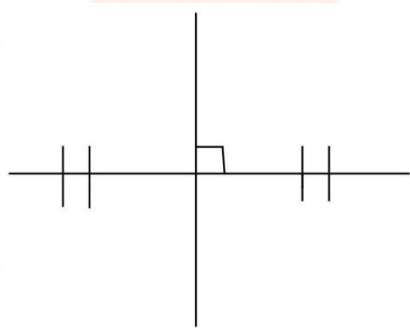
$$\tan \theta = \frac{1}{\sqrt{3}}$$

$$\tan \theta = 30^\circ$$

Hence, the angle of value is $\tan \theta = 30^\circ$.

Question.6 Write the locus of the points equidistant from the two given points.

Solution.



The two points makes are line segment of perpendicular equidistant.

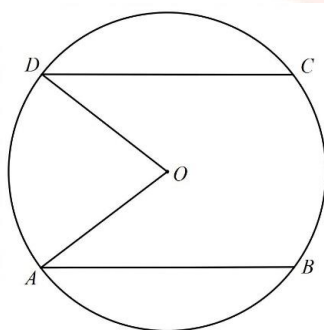
Then,

$$Ax + By + C = 0$$

$$P(x_1, y_1)(x_2, y_2)$$

Question.7 Write the ratio between the chord which are equidistant from the centre of a circle.

Solution.



Two chords AB and CD of a circle with center O which are equal distance from the center $OE=OF$,

Where,

Join OA and OC

$OE \perp AB$ and $OF \perp CD$

Since the perpendicular from the center of a circle to a chord bisector.

$AE = EB$ and $CF=FD$

Therefore,

$\angle AEO = \angle CFO$ (each equal 90°)

$OE = OF$

Thus, the chord ratio is equal.

Then AB: CD: 1:1

The chord ratio is 1:1

Hence, the chord ratio is 1:1

Question.8 A dice is thrown once. Find the probability of getting an odd number.

Solution.

Simplify the expression,

In a single throw of a die, all possible outcomes are 1, 2, 3, 4, 5, 6

Total number of possible outcomes = 6

Then, the favourable outcomes are 1, 3, 5

Number of favourable outcomes = 3

Probability (getting a odd number) $\Rightarrow \frac{3}{6} = \frac{1}{2}$

Hence, the Probability (getting a odd number) $= \frac{1}{2}$

Question.9 In a city, the fare of a taxi for first kilometer is Rs 5 and after Rs 3. If distance covered is x km and fare is Rs y , then express in the form of equation.

Solution.

Simplify the equation,

Total distance covered = x

Fare for 1st kilometer = 5

Fare for the rest of the distance = $Rs(x-1)3$

Total fare

$$= Rs[5 + (x-1)3]$$

$$y = 3x + 2$$

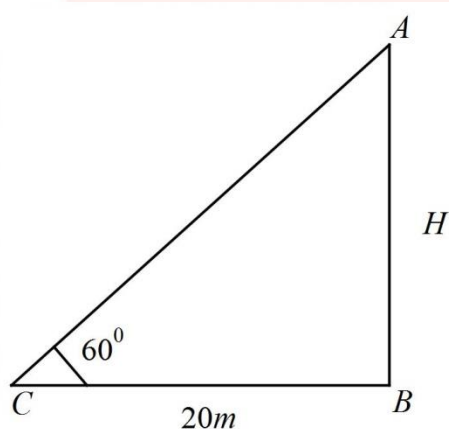
$$3x - y + 2 = 0$$

Hence, the value of equation $3x - y + 2 = 0$.

Question.10 If the elevation angle of a camera situated at the top of a pole from a point 20 metre away for the base of the pole is 60° , find the height of the pole.

Solution.

Simplify the expression,



$$\frac{P}{H} = \tan 60^\circ$$

$$\frac{h}{20} = \sqrt{3}$$

$$h = 20\sqrt{3}$$

$$h = 20\sqrt{3}$$

Hence, the height of pole is $h = 20\sqrt{3}$.

PART-B

Question.11 Find the square root of 6889 by using Dwandwa Yoga Method.

Solution.

Simplify the expression,

Using the dwandwa yoga method,

8	6 8	8 9
16		4 0
	8	3

Step:1 first square root = 8 and remainder $\Rightarrow 68 - 64 = 4$

$$8 \times 2 = 16$$

Step:2 new dividend $48 \div 16 = 3$ remainder = 0

Implement dividend $9 - 3^2 = 0$

Hence, the value is 83.

Question.12 If the product of two numbers is 525 and their H.C.F is 5, then find their L.CM.

Solution.

Simplify the expression,

$$\text{HCF (A,B)} \times \text{LCM(A,B)} = \text{Product of A and B.}$$

$$LCM \times 5 = 525$$

$$LCM = \frac{525}{5}$$

$$LCM = 105$$

Hence, the value of $LCM = 105$.

Question.13 The total surface area of a cube is 216 square meter. Find the side of cube.

Solution.

Simplify the expression,

We know that, the surface area of cube is $6a^2$

Then,

$$6a^2 = 216$$

$$a^2 = \frac{216}{6}$$

$$a^2 = 36$$

$$a = 6$$

Hence, the side of cube is $a = 6$.

Question.14 The radius of semi-sphere is 7 cm, find the total surface area of it.

Solution.

Simplify the expression,

We know that the semi-sphere total surface area is $3\pi r^2$

Radius $r = 7$

$$= 3\pi r^2$$

$$= 3 \cdot \frac{22}{7} \cdot (7)^2$$

$$= 3 \times \frac{22}{7} \times 7 \times 7$$

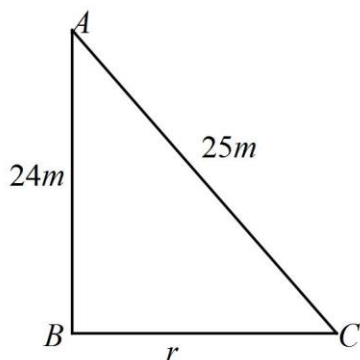
$$= 3 \times 22 \times 7$$

$$= 462\text{cm}^2$$

Hence, the value of total hemi-sphere of area is 462cm^2 .

Question.15 A CCTV camera is placed on the top of a 24m high pole in such a way that traffic can be seen beyond 25 meter of line of sight of it. Find the area of the Green patch around the pole.

Solution.



Using Pythagoras theorem

$$(25)^2 = (24)^2 + r^2$$

$$r^2 = 625 - 576$$

$$r^2 = 49$$

$$r = 7$$

Invisible area of circle $= \pi r^2$

$$\Rightarrow \frac{22}{7} \times 7 \times 7$$

$$\Rightarrow 22 \times 7$$

$$\Rightarrow 154m^2$$

PART-C

Question.16 By using division algorithm method find quotient and remainder when polynomial $P(x) = x^4 - 3x^2 + 4x - 3$ is divided by $g(x) = x^2 + 1 - x$.

Solution.

Simplify the expression,

First we write the given polynomials in standard form in decreasing order of degrees and then perform the division as shown below.

$$\begin{array}{r}
 x^2 + x^2 - 3 \\
 x^2 + 1 - x \overline{) x^4 - 3x^2 + 4x - 3} \\
 \underline{x^4 + x^2 \dots\dots\dots x^3} \\
 - \quad - \\
 \hline
 x^3 - 4x^2 + 4x - 3 \\
 \underline{x^3 - x^2 + x} \\
 - \quad + \quad - \\
 \hline
 3x^2 + 3x - 3 \\
 \underline{3x^2 + 3x - 3} \\
 - \quad - \quad + \\
 \hline
 0
 \end{array}$$

$$\begin{aligned}
 p(x) &= g(x) \times q(x) + r(x) \\
 x^4 - 3x^2 + 4x - 3 &= (x^2 - x + 1)(x^2 + x - 3) + 0 \\
 &= x^4 - x^3 + x^2 + x^3 - x^2 + x - 3x^2 + 3x - 3 \\
 &= x^4 - 3x^2 + 4x - 3
 \end{aligned}$$

Hence, $x^4 - 3x^2 + 4x - 3$ and remainder 0.

Question.17 If second and third of an Arithmetic Progression are 3 and 5 respectively, then find the sum of first 20 term of it.

Solution

Simplify the A.P expression,

Second and third terms of an Arithmetic Progression are 3 and 5,

$$\begin{aligned}
 \text{Here, } a &= 3, \quad d = a_3 - a_2 = (5 - 3) = 2 \quad \text{and } n = 20 \\
 d &= a_2 - a_1 \Rightarrow 2 = 3 - a_1 \Rightarrow a_1 = 1
 \end{aligned}$$

We using the formula $S_n = \frac{n}{2} \cdot [2a + (n-1)d]$

$$S_n = \frac{20}{2} \cdot [1 \cdot 2 + (20-1)2]$$

$$S_{20} = 10[2 + (19) \times 2]$$

$$S_{20} = 10(40)$$

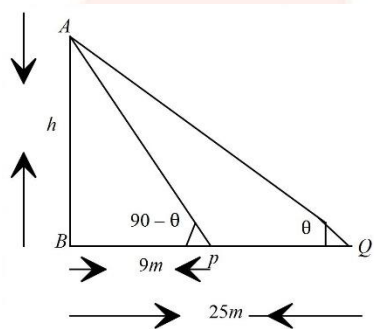
$$S_{20} = 400$$

Hence, the value is $S_{20} = 440$.

Question.18 The angles of elevation of the top of a tower from two points at a distance of 9m.

And 25 m from the base of the tower in the same straight line are complementary .find the height of the tower.

Solution.



Simplify the expression,

$\triangle ABP$

$$\frac{h}{9} = \tan \theta$$

$$\frac{h}{9} = \tan (90 - \theta) = \cot \theta \quad [\theta = (90 - \theta)]$$

$\triangle ABQ$

$$\frac{h}{25} = \tan \theta \dots\dots(ii)$$

Equation (i) and (ii) of multiply by

$$\frac{h}{9} \times \frac{h}{25} = \cot \theta \times \tan \theta$$

$$\frac{h^2}{9 \times 25} = 1$$

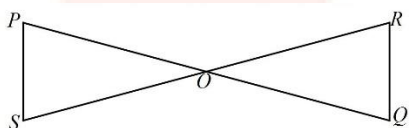
$$h^2 = 9 \times 25$$

$$h = 3 \times 5$$

$$h = 15m$$

Hence, the height of tower is $h = 15m$.

Question.19 In given figure if $OP \cdot OQ = OR \cdot OS$, then show that $\angle OPS = \angle ORQ$ and $\angle OQR = \angle OSP$.



Solution.

Prove the expression,

$$OP \cdot OQ = OR \cdot OS$$

$$\frac{OP}{OS} = \frac{OR}{OQ}$$

And

$$\angle POQ = \angle ROQ \quad [\quad]$$

then,

$$\triangle POQ \sim \triangle ROQ$$

$$\angle OPQ = \angle ORQ$$

and

$$\angle OQP = \angle OQR$$

Hence, the prove it.

Question.20 In a triangle ABC, the median AD, BE and CF pass through the point G. If $AD = 9\text{cm}$, $GE = 4.2\text{cm}$, $GC = 6\text{cm}$, then find the values of lengths of AG, BE and FG.

Solution.

Simplify the expression,

$\angle ABD$

$AG : GD :: 2 : 1$

given in $AD = 9\text{cm}$

$$AG = \frac{2}{3} \times 9$$

$$AG = 2 \times 3 = 6\text{cm}$$

$\angle BEC$

$BG : GE :: 2 : 1$

given in $GE = 4.2$

$$GE = \frac{4.2}{1} = 4.2$$

$$BE = 3 \times 4.2 = 12.6\text{cm}$$

$\angle BEC$

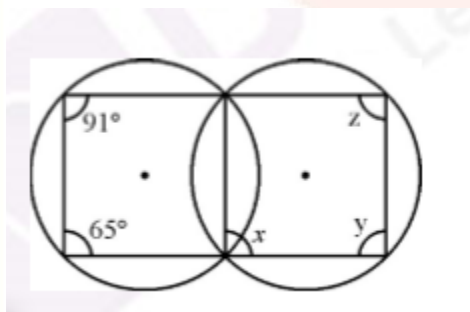
$FG : GC :: 2 : 1$

$$FG = \frac{2}{3} \times 6 = 4\text{cm}$$

$$FG = 4\text{cm}$$

Hence, the value is $AG = 6\text{cm}$, $BE = 12.6$, and $FG = 4\text{cm}$.

Question.21 In the given figure some angles are represented by x, y and z. Find the values of these angles.



Solution.

Simplify the expression,

$$91^\circ + x_1 = 180^\circ$$

$$x_1 = 180^\circ - 91^\circ$$

$$x_1 = 89^\circ$$

we know that (A straight line equal to 180°)

$$x_1 + x = 180^\circ$$

$$89^\circ + x = 180^\circ$$

$$x = 180^\circ - 89^\circ$$

$$x = 91^\circ$$

simplify further,

$$x + z = 180^\circ$$

$$91^\circ + z = 180^\circ$$

$$z = 89^\circ$$

then,

$$y_1 = 91^\circ$$

$$y_1 + y = 180^\circ$$

$$91^\circ + y = 180^\circ$$

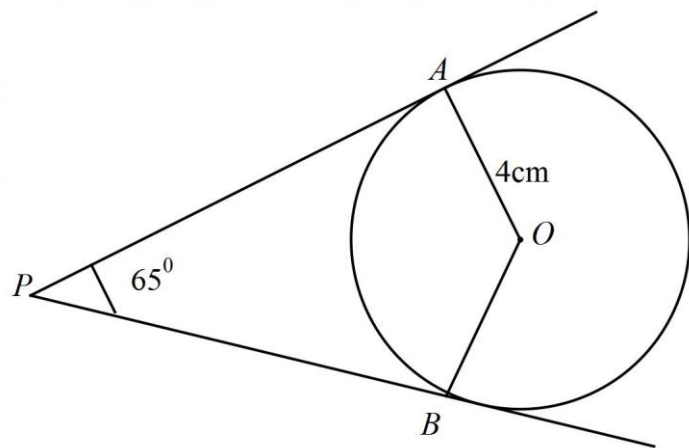
$$y = 180^\circ - 91^\circ = 89^\circ$$

Hence, the value is $x = 91^\circ$, $y = 89^\circ$, $z = 89^\circ$.

Question.22 Draw two tangents PA and PB from an external point P, to a circle of radius 4 cm, where angle between PA and PB is 65° .

Solution.

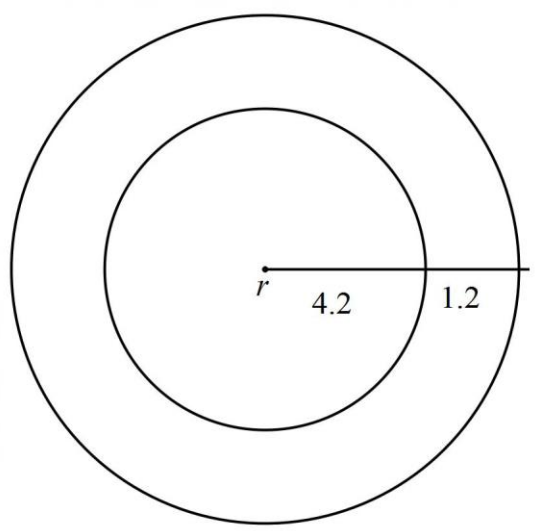
Draw the figure below in,



Hence, draw above in figure.

Question.23 The radius of a circular park is 4.2 m. A path of 1.4 m width is made around the circular park. Find the area of the path.

Solution.



Simplify the expression,

Radius of immaculate 4.2m

And width external circular = $4.2 + 1.4 = 5.6 = R$

Area of path,

$$\begin{aligned}
 &= \pi(R^2 - r^2) \\
 &= \pi(R + r)(R - r) \\
 &= \pi(5.6 + 1.4)(5.6 - 1.4) \\
 &= \frac{22}{7} \times 7 \times 4.2 \\
 &= 22 \times 4.2 = 92.4 \text{ cm}^2
 \end{aligned}$$

Hence, the area of path is 92.4 cm^2 .

Question.25 In a bag one white ball, two black balls and three red balls of same size are placed. A ball is drawn at random from this bag. Find the probability.

- (i) Ball is white
- (ii) Ball is not black
- (iii) Ball is red

Solution.

Simplify the probability,

Total number of all possible outcomes $\Rightarrow (1 + 2 + 3 = 6)$.

- (i) Number of white ball = 1
Probability (getting a white ball) $= \frac{1}{6}$
- (ii) Number is not black ball $(1 + 3 = 4)$
Probability (getting not black ball) $= \frac{4}{6} = \frac{2}{3}$
- (iii) Number of red ball = 3
Probability (getting a red ball) $= \frac{3}{6} = \frac{1}{2}$

Hence, the probability value are $\frac{1}{6}$, $\frac{2}{3}$ and $\frac{1}{2}$.

PART-D

Question.26

Question.27 (i) Prove that:

$$\sqrt{\frac{1+\cos\theta}{1-\cos\theta}} = \operatorname{cosec}\theta + \cot\theta$$

Solution.

Prove the expression,

$$\begin{aligned} & \sqrt{\frac{1+\cos\theta}{1-\cos\theta}} \times \sqrt{\frac{1+\cos\theta}{1-\cos\theta}} \\ &= \frac{(1+\cos\theta)}{\sqrt{(1+\cos\theta)(1-\cos\theta)}} = \frac{(1+\cos\theta)}{\sqrt{(1-\cos\theta)^2}} \\ &= \frac{(1+\cos\theta)}{\sqrt{\sin^2\theta}} = \frac{(1+\cos\theta)}{\sin\theta} \\ &= \frac{1}{\sin\theta} + \frac{\cos\theta}{\sin\theta} = (\operatorname{cosec}\theta + \cot\theta) \end{aligned}$$

Hence, the prove it.

(ii) Prove:

$$\frac{\tan\theta}{1-\cot\theta} + \frac{\cot\theta}{1-\tan\theta} = 1 + \tan\theta + \cot\theta$$

Taking LHS

$$\frac{\tan\theta}{1-\cot\theta} + \frac{\cot\theta}{1-\tan\theta}$$

$$\begin{aligned}
 & \frac{\tan \theta}{1 - \frac{1}{\tan \theta}} + \frac{1}{1 - \tan \theta} \\
 &= \frac{\tan \theta}{\frac{\tan \theta - 1}{\tan \theta}} - \frac{1}{\tan \theta(1 - \tan \theta)} \\
 &= \frac{\tan^2 \theta}{\tan \theta - 1} - \frac{1}{\tan \theta(\tan \theta - 1)} \\
 &= \frac{\tan^3 \theta - 1}{\tan \theta(\tan \theta - 1)} \\
 &= \frac{(\tan \theta - 1)(\tan^2 \theta + \tan \theta + 1)}{\tan \theta(\tan \theta - 1)} \\
 &= \frac{\tan^2 \theta + \tan \theta + 1}{\tan \theta} \\
 &= \tan \theta + 1 + \cot \theta \Rightarrow RHS
 \end{aligned}$$

Hence, prove it.

OR

- (i) If $\sin \theta + \cos \theta = p$ and $\sec \theta + \operatorname{cosec} \theta = q$, then prove that $q(p^2 - 1) = 2p$

Solution.

Prove the expression,

$$(\sin \theta + \cos \theta)^2 = p^2$$

$$\sin^2 \theta + \cos^2 \theta + 2 \sin \theta \cos \theta = p^2$$

$$p^2 = \sin^2 \theta + \cos^2 \theta + 2 \sin \theta \cos \theta$$

$$p^2 = 1 + 2 \sin \theta \cos \theta$$

$$\sec \theta + \operatorname{cosec} \theta = q$$

$$\frac{1}{\cos \theta} + \frac{1}{\sin \theta} = q$$

$$q = \frac{\sin \theta + \cos \theta}{\sin \theta \cos \theta}$$

Putting the value of p and q in equation $q(p^2 - 1)$.

$$\frac{\sin \theta + \cos \theta}{\sin \theta \cos \theta} (1 + 2 \sin \theta \cos \theta - 1)$$

$$= 2(\sin \theta + \cos \theta) = 2p$$

Hence, prove it.

(ii) Prove that:

$$\frac{\cos A}{1 - \tan A} + \frac{\sin A}{1 - \cot A} = \sin A + \cos A$$

Taking LHS

$$= \frac{\cos A}{1 - \tan A} + \frac{\sin A}{1 - \cot A}$$

$$= \frac{\cos A}{\left(1 - \frac{\sin A}{\cos A}\right)} + \frac{\sin A}{\left(1 - \frac{\cos A}{\sin A}\right)}$$

$$= \frac{\cos^2 A}{(\cos A - \sin A)} + \frac{\sin^2 A}{(\sin A - \cos A)}$$

$$\Rightarrow \frac{\cos^2 A - \sin^2 A}{(\cos A - \sin A)} = \frac{(\cos A - \sin A)(\cos A + \sin A)}{(\cos A - \sin A)}$$

$$\Rightarrow (\cos A + \sin A) = RHS$$

Hence, prove it.

Question.28 (i) If distance between points $(x, 3)$ and $(5, 7)$, then find the value of x .

(ii) Find the ratio in which the line $3x + y = 9$ divides the line segment joining the points $(1, 3)$ and $(2, 7)$.

Solution

(i) Simplify the expression,

We know that,

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$(x, 3), (5, 7)$

$$5 = \sqrt{(5-x)^2 + (7-3)^2}$$

$$25 = (5-x)^2 + (4)^2$$

$$25 = 25 + x^2 - 10x + 16$$

$$x^2 - 10x + 16 = 0$$

$$x^2 - 8x - 2x + 16 = 0$$

$$x(x-8) - 2(x-8) = 0$$

$$(x-2)(x-8)$$

$$x = 2$$

$$x = 8$$

$$(2, 8)$$

(ii) Given points are (1, 3) and (2, 7)

$$x = \frac{2k+1}{k+1}$$

$$y = \frac{7k+3}{k+1}$$

$$= 3\left(\frac{2k+1}{k+1}\right) + \left(\frac{7k+3}{k+1}\right) = 9$$

$$\Rightarrow \frac{6k+3+7k+3}{k+1} = 9$$

$$\Rightarrow 13k+6 = 9k+9$$

$$\Rightarrow 4k = 3$$

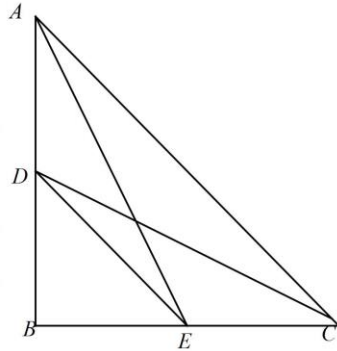
$$\Rightarrow k = \frac{3}{4}$$

$$\Rightarrow 3:4$$

Hence, the ratio is 3:4.

Question.29 ABC is right angled triangle whose $\angle B$ is right angle. If points D and E are situated on the sides AB and BC respectively, then prove that $AE^2 + CD^2 = AC^2 + DE^2$.

Solution.



In

$\triangle ABE$ (right triangle)

$$AE^2 = AB^2 + BE^2 \dots\dots(i)$$

In $\triangle BCD$ (right triangle)

$$CD^2 = BD^2 + BC^2$$

$$AE^2 + CD^2 = AB^2 + BE^2 + BD^2 + BC^2$$

$$= AB^2 + BC^2 + BD^2 + BE^2$$

In $\triangle ABC$

$$AC^2 = AB^2 + BC^2$$

In $\triangle BED$

$$DE^2 = BD^2 + BE^2$$

$$AE^2 + CD^2 = AB^2 + BE^2 + BD^2 + BC^2$$

$$AE^2 + CD^2 = AC^2 + DE^2$$

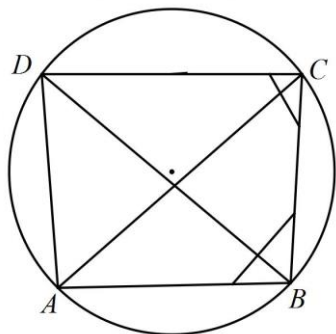
Hence, prove it.

OR

If two sides of a cyclic quadrilateral are parallel, then prove that other sides are equal and its diagonals are also equal to each other.

Solution.

Prove the expression,



Given – $AB \parallel CD$,

$ABCD$ is a cyclic equal.

To prove - $AD=BC$, $AC=BD$

Proof - $\angle B + \angle D = 180^\circ$ (1)

$\angle B + \angle C = 180^\circ$ (2)(interior angle of $\parallel AB \& CD$)

(1)=(2)

$\angle A + \angle D = \angle B + \angle C$

$\angle C = \angle D$

In $\triangle ADC$ & $\triangle BDC$

$CD = CD$ (common)

$\angle C = \angle D$

$\angle CAD = \angle CBD$

By AAS Rule,

$\triangle ADC \cong \triangle BDC = AD=BC$,

- $AC = BD$

Hence, prove it.

Question.30 Find the mean and mode of the following frequency distribution.

Score	20-30	30-40	40-50	50-60	60-70
Number of students	4	28	42	20	6

Solution.

Simplify the modal expression,

Score	Number of students f_i	Midvalue x_i	$u_i = \frac{(x_i - A)}{h}$ $= \frac{(x_i - 100)}{10}$	$(f_i \times u_i)$
20-30	4	25	-7.5	-30
30-40	28	35	-6.5	-182
40-50	42	45	-5.5	-231
50-60	20	55	-4.5	-90
60-70	6	65	-3.5	-21
	$\sum f_i = 100$			$\sum (f_i \times u_i) = -554$

$$A = 45, h = 10, \sum f_i = 100 \text{ and } \sum f_i \times u_i$$

$$\bar{x} = \left\{ h + \frac{\sum (f_i \times x_i)}{\sum f_i} \right\}$$

$$= 45 + \left\{ 10 \times \frac{554}{100} \right\} = 45 + 55.4 = 100.4$$

Hence, the mean is 100.4

As the life time (in hours) 40-50 has maximum frequency, so it is the modal life time in hours.

$$x_k = 40, h=10, f_k = 42, f_{k-1} = 28 \text{ and } f_{k+1} = 20$$

$$\text{Mode, } M_0 = x_k + \left\{ h \times \frac{(f_k - f_{k-1})}{(2f_k - f_{k-1} - f_{k+1})} \right\}$$

$$= 40 + \left\{ 10 \times \frac{(42 - 28)}{(2 \times 42 - 28 - 20)} \right\}$$

$$= 40 + \left\{ 10 \times \frac{14}{(84 - 48)} \right\}$$

$$= 40 + \left\{ 20 \times \frac{14}{36} \right\}$$

$$= 40 + (7.77)$$

$$= 47.77$$

Hence, the mode is 47.77.